



**GURUNANAK INSTITUTE OF TECHNOLOGY**  
(AN UGC AUTONOMOUS)

Approved by AICTE, and Affiliated with JNTU Hyderabad  
( Ibrahimpatnam, R.R. Dist., Telangana,501506)

**MINOR PROJECT PRESENTATION**  
**ON**  
**SIMPLE FITNESS WEB APP USING OPENCV**

**UNDER THE GUIDANCE OF:**

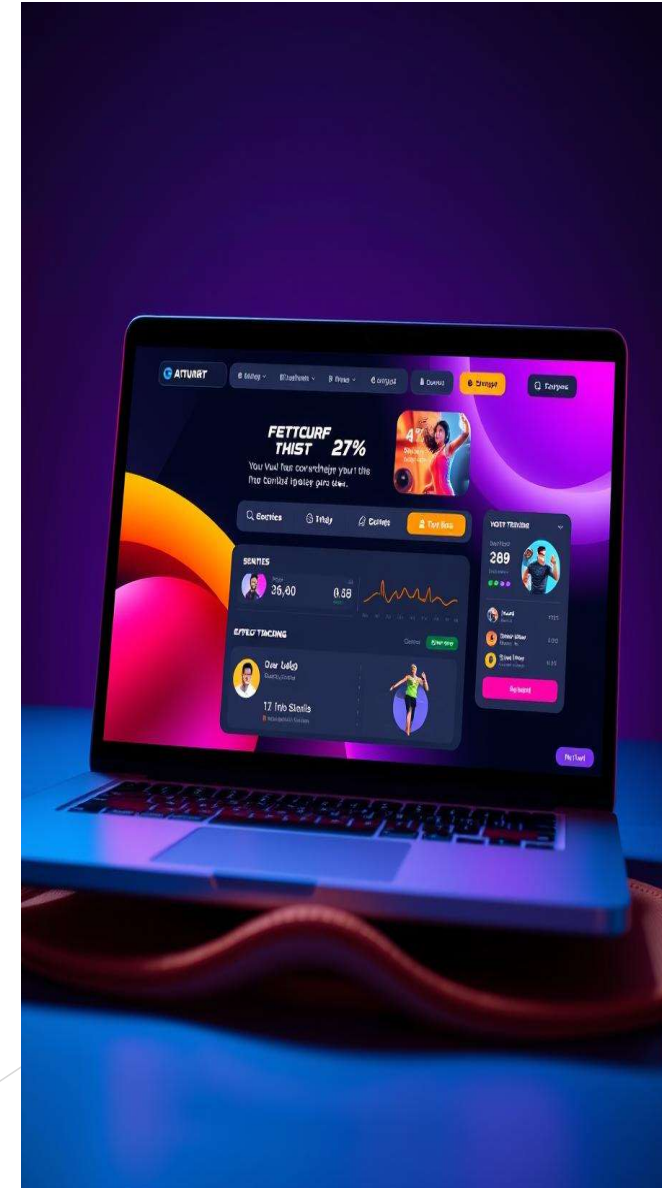
Mrs. K. NIRMALA  
PROJECT GUIDE

**Team Members**

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# Introduction

- FitHub revolutionizes fitness tracking and analysis with its innovative integration of OpenCV and MediaPipe technologies.
- This web application offers a holistic approach to fitness monitoring by harnessing OpenCV's powerful image processing capabilities for real-time video analysis of exercises like push-ups, squats, and lunges.
- Through precise repetition counting and instant feedback on form and technique, users can optimize their workout routines for maximum effectiveness.
- MediaPipe's advanced pose estimation models further enhance the app's functionality by accurately detecting key body landmarks and tracking movement dynamics.



## **ABSTRACT**

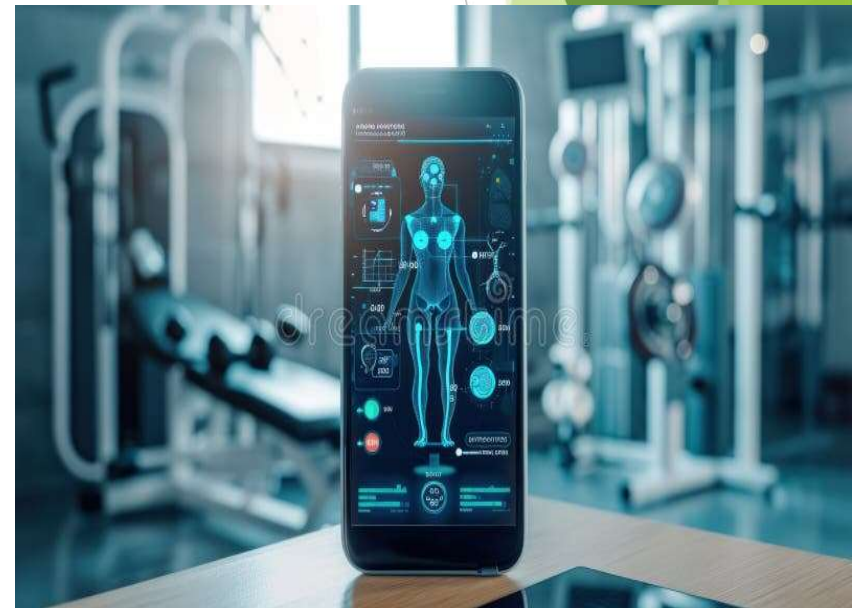
The web application integrates OpenCV and MediaPipe to provide a comprehensive fitness tracking and analysis platform. Leveraging OpenCV's robust image processing capabilities, the app offers real-time video analysis for exercises such as push-ups, squats, and lunges, accurately counting repetitions and providing instant feedback on form and technique. MediaPipe's pose estimation models enhance the app's by precisely detecting key body landmarks and tracking movement dynamics. This combined technology enables users to receive personalized workout progress tracking over time. Fit Hub empowers users to achieve their fitness goals effectively by leveraging cutting-edge computer vision techniques in a user-friendly and accessible application.

## EXISTING SYSTEM:

- Some software tools existed before OpenCV, such as MATLAB's Image Processing Toolbox and Intel's Image Processing Library (IPP).
- These tools provided a range of image processing functions and algorithms but were often proprietary, expensive, and required specific programming languages or environments.
- Machine vision systems were also prevalent in industrial settings, employing cameras, lighting, and custom software to automate tasks like quality inspection, object recognition, and robotic guidance.

## EXISTING SYSTEM DISADVANTAGES:

- It needed high processing time.
- MATLAB's Image Processing Toolbox can be limited in terms of speed and efficiency for large-scale image processing tasks.



## **PROPOSED SYSTEM :**

- OpenCV (Open Source Computer Vision Library) is a crucial tool in the field of computer vision and image processing, offering significant advantages across various industries.
- Its open-source nature makes it freely available, promoting widespread access and collaboration among researchers, developers, and hobbyists.
- The library provides a comprehensive set of tools and algorithms for tasks such as image recognition, object detection, image stitching, and 3D reconstruction,

### **PROPOSED SYSTEM ADVANTAGES:**

- OpenCV offers a wide range of pre-built computer vision algorithms and functions, saving developers time in implementing complex image processing tasks.
- Its cross-platform support allows OpenCV to be used on various operating systems and hardware configurations, enhancing its versatility and accessibility.

# FIT HUB

- Fit Hub is a comprehensive fitness tracking and analysis platform that integrates cutting-edge computer vision technologies, including OpenCV and MediaPipe.
- This web application offers real-time video analysis for exercises such as push-ups, squats, and lunges, accurately counting repetitions and providing instant feedback on form and technique.
- MediaPipe's pose estimation models enhance the app's capabilities by precisely detecting key body landmarks and tracking movement dynamics.

Welcome to our AI-based yoga trainer website, where technology meets mindfulness to help you achieve your fitness goals.



## Yoga Guruji

Experience yoga like never before with our app's revolutionary computer vision technology. Seamlessly follow along with guided routines as our app effortlessly recognizes and classifies various yoga poses. Whether you're a novice or a seasoned practitioner, personalized feedback and guidance await you. Our standout feature? Real-time pose correction. By analyzing your movements and aligning them with ideal form, we safeguard against potential injuries and maximize the benefits of each posture.



## Exercise Trainer

Our app counts the number of times you complete each exercise, so you can easily track your progress over time. Whether you're aiming to do more reps, lift heavier weights, or simply maintain your current fitness level, our repetition tracking feature makes it easy to monitor your progress and stay motivated. That's why our app provides real-time feedback on your form as you exercise. Our app analyzes your movements and corrects your form if you're doing an exercise incorrectly. With our form correction feature

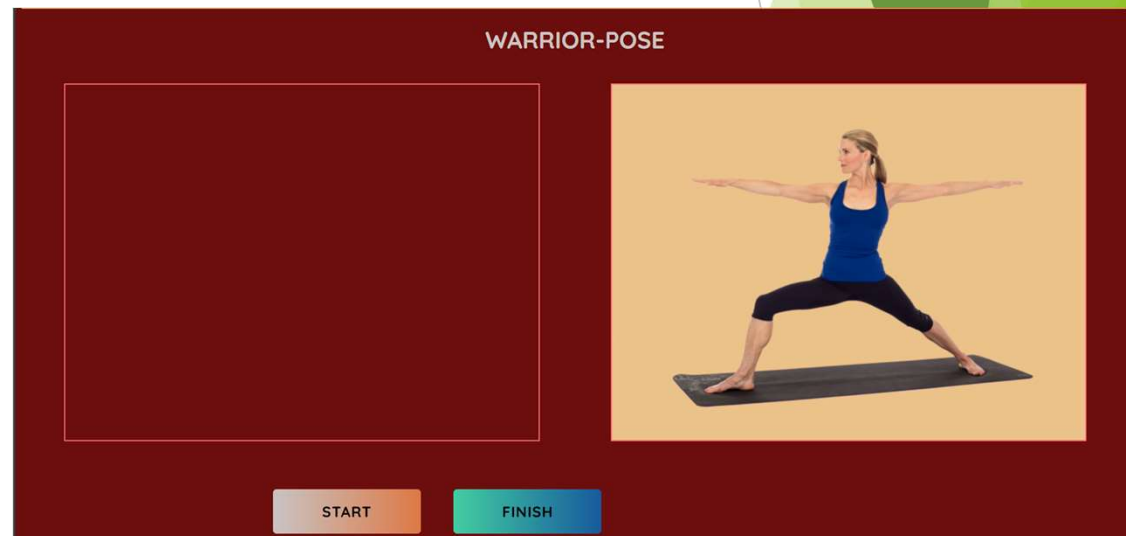
## MODULES DESCRIPTION:

### **OpenCV:**

Utilized for robust image processing capabilities, enabling real-time video analysis for exercises and accurate repetition counting.

**MediaPipe:** Integrated for advanced pose estimation models, enhancing the app's functionality in detecting key body landmarks and tracking movement dynamics.

**Fitness Tracking Module:** Manages workout data, including exercise types, repetitions, and user progress tracking over time.



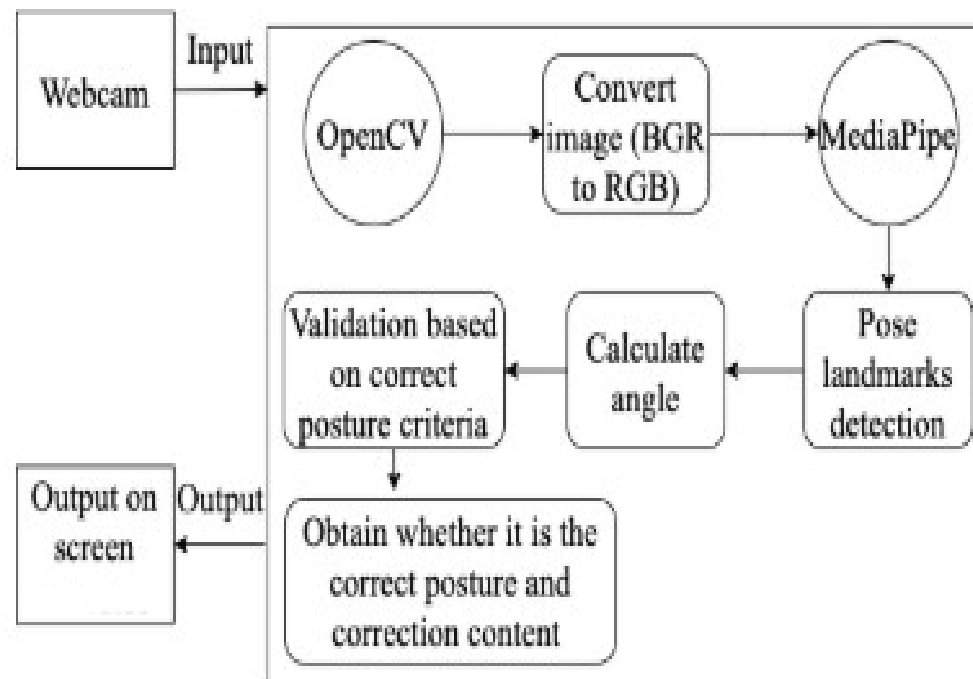


- **Feedback Module:** Provides instant feedback on form and technique during exercises, leveraging data from OpenCV and MediaPipe.
- **User Interface:** Designs a user-friendly interface for easy navigation, exercise tracking, and accessing personalized workout insights.





# System Architecture



# REQUIREMENTS

## HARDWARE REQUIREMENTS

- Processor : dual core 2 duos.
- RAM : 4gb DD RAM
- Hard disk : 250 GB

## SOFTWARE REQUIREMENTS

- Operating System : Windows 7/8/10
- Platform : Spyder3
- Programming Language : Python
- Front End : HTML, CSS

# Python

- Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable.

## Features of Python

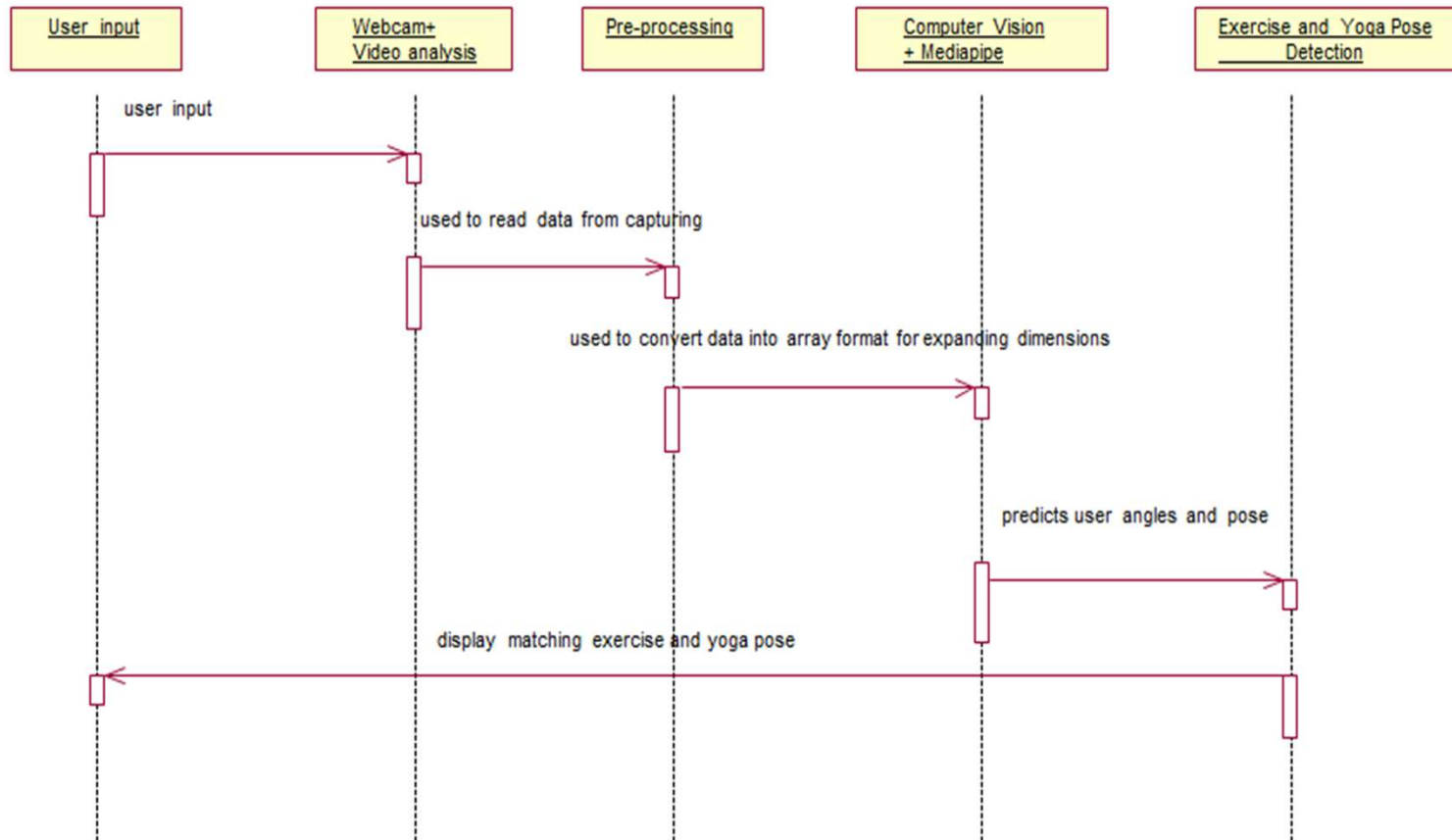
- **Easy-to-maintain** – Python's source code is fairly easy-to-maintain.
- **A broad standard library** – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode** – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable** – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.

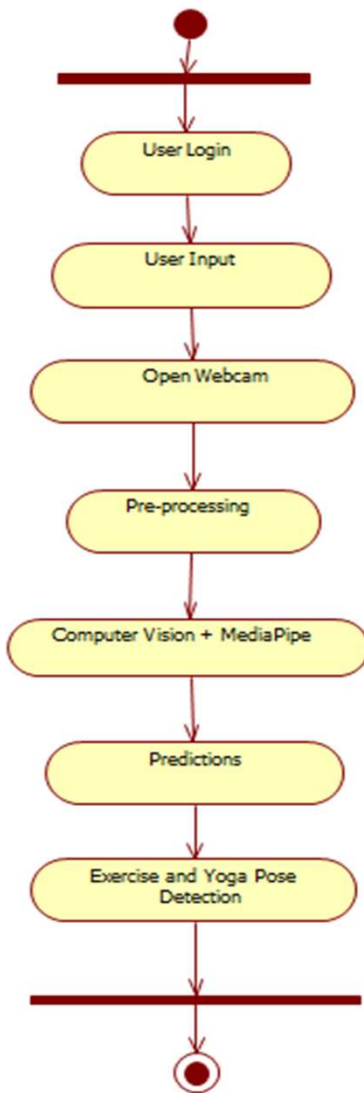
## Libraries used in python

- NumPy - mainly useful for its N-dimensional array objects.
- pandas - Python data analysis library, including structures such as data frames.
- matplotlib - 2D plotting library producing publication quality figures.
- scikit-learn - the machine learning algorithms used for data analysis and data mining tasks.



# Sequence diagram



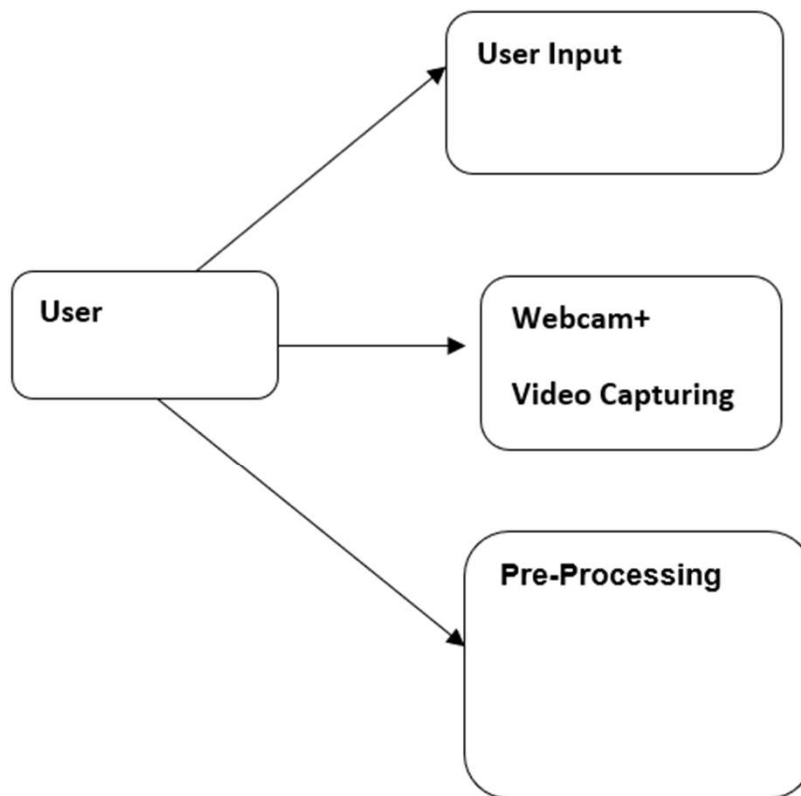


## ACTIVITY DIAGRAM

- Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency.
- In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

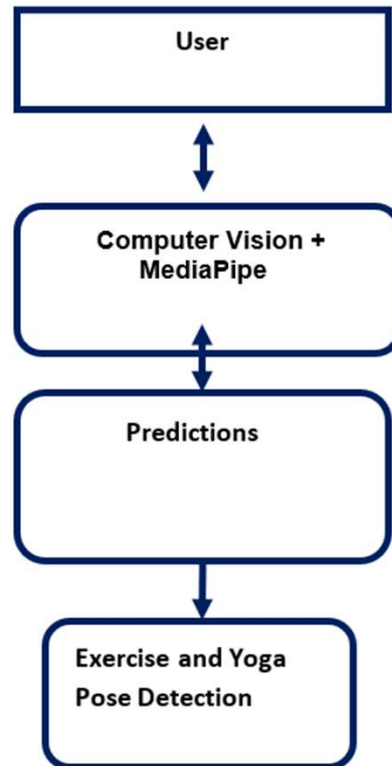
## Data Flow Diagram

Level 0:





## Level 1:



## SOFTWARE TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product.

- **Unit testing**

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program input produce valid outputs.

- **Functional testing**

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

- **System testing**

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results.

- **Integration Testing**

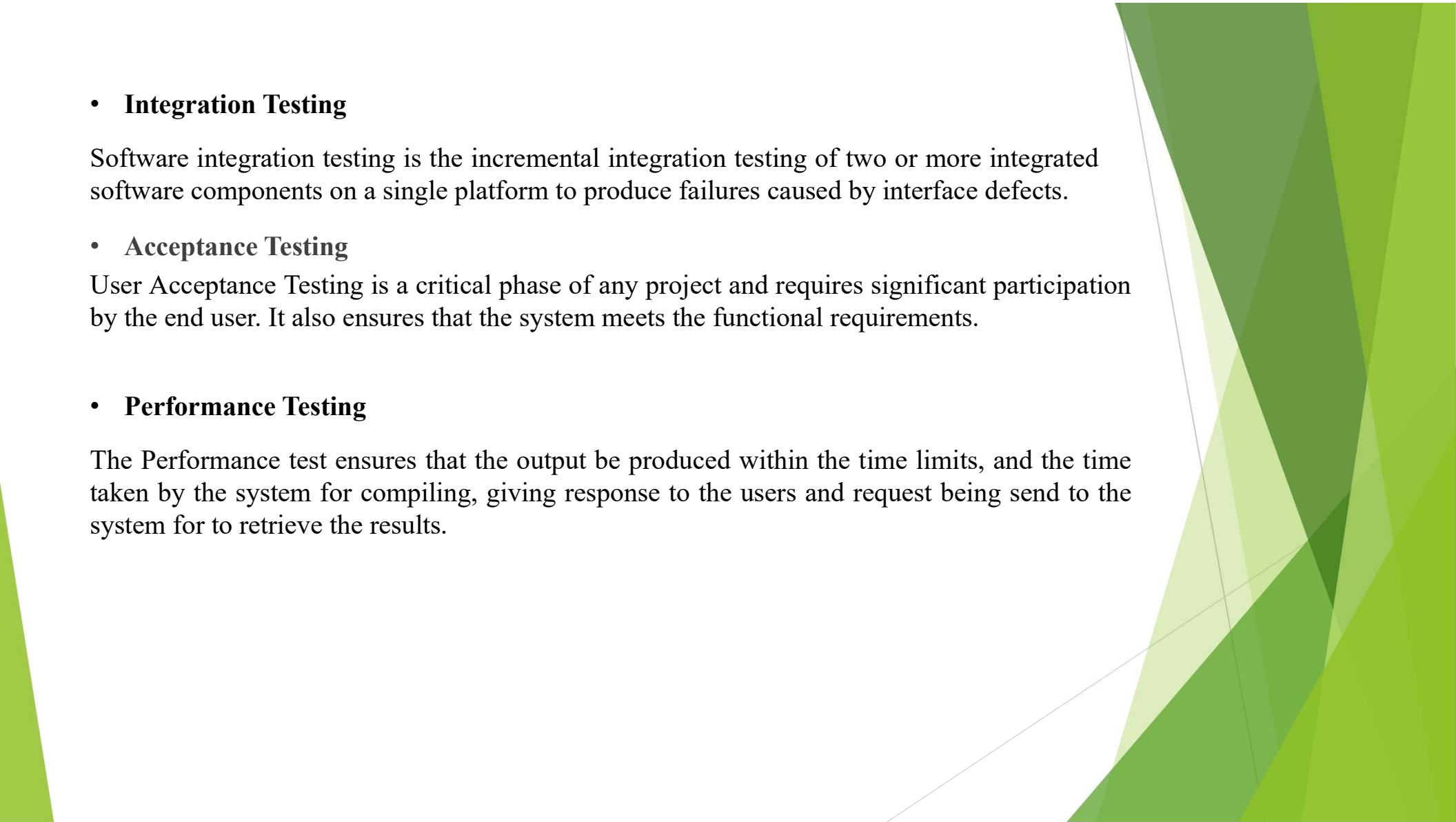
Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

- **Acceptance Testing**

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

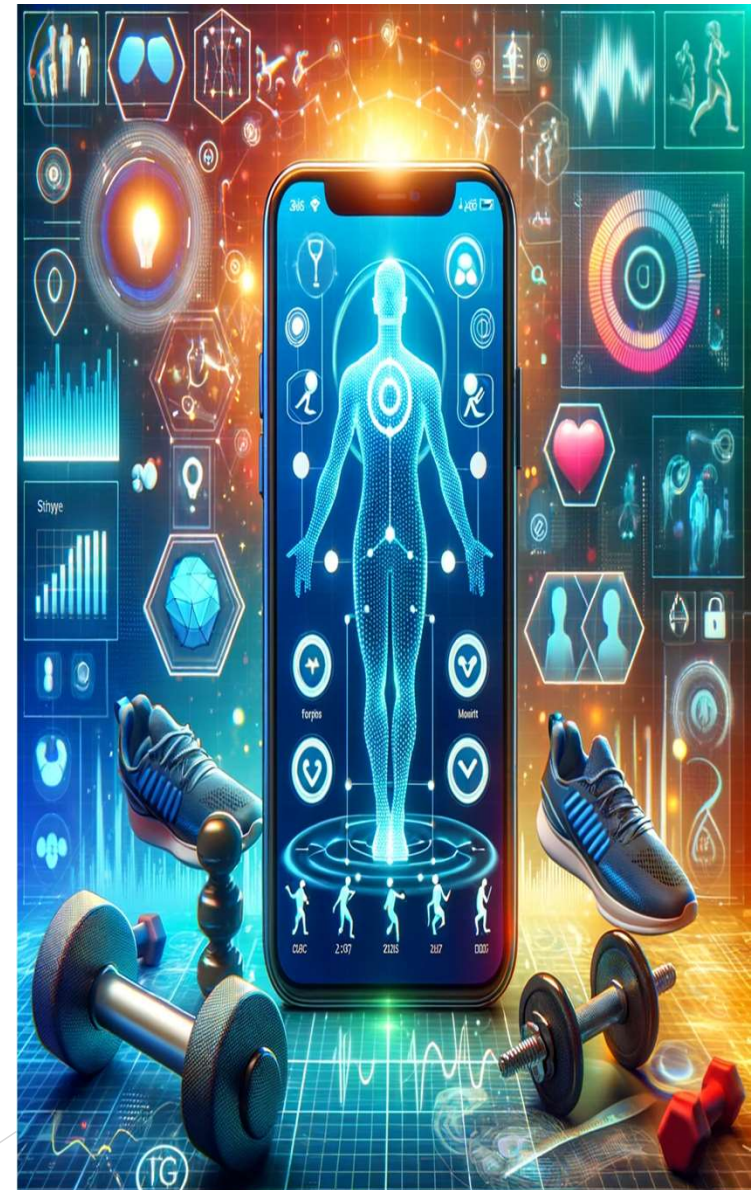
- **Performance Testing**

The Performance test ensures that the output be produced within the time limits, and the time taken by the system for compiling, giving response to the users and request being send to the system for to retrieve the results.



## FUTURE ENHANCEMENTS

- **Enhanced feedback mechanisms**, including real-time posture correction and interactive virtual trainers, will elevate user guidance and engagement.
- **Integration with virtual reality (VR)** technology promises immersive workout experiences and diverse training environments.
- **Social integration** features like user communities and challenges will foster motivation and accountability among users.
- The development of a **mobile app version** with offline capabilities and mobile-friendly interfaces will extend FitHub's accessibility and convenience.
- **Nutrition tracking, health monitoring**, and AI-driven insights will provide holistic wellness support alongside fitness activities.



## CONCLUSION

- In conclusion, FitHub represents the future of fitness tracking and analysis, combining cutting-edge technologies with user-centric design to offer a holistic and personalized fitness experience.
- With advancements such as machine learning integration, enhanced feedback mechanisms, virtual reality immersion, and social integration.
- FitHub empowers users with actionable data, engaging workouts, and a supportive community environment.

THANK YOU

